MIL-B-36636A
18 April 1973
SUPERSEDING
MIL-B-36636
22 January 1970

MILITARY SPECIFICATION

BONE PLATE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers curved bone plates of one-piece construction, used in the internal fixation of bone segments.

1.2 Classification. Bone plates shall be of the following sizes and styles as specified (see 6.2).

Size	Overall Length (inches)
1	2.25
2	2.8
3	3.35
4	3.9
5	4.7
6	5.5
7	6.3
8	7.1
9	7.9

Style	Number of Holes
1	4
2	6
3	8

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

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SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up.
PPP-T-76	Tape, Pressure Sensitive Adhesive Paper, (for Carton Sealing).

Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-151	Metals, Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

Defense Personnel Support Center (DPSC)

1826	Bone Plate.

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM	F55	Stainless Steel Bars and Wire for Surgical Implants.
ASTM	E45	Determining the Inclusion Content of Steel
ASTM	F86	Recommended Practice for Surface Requirements of Metallic Surgical Implants

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of American (U.S.P.)

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042).

3. REQUIREMENTS

3.1 Material. The bone plate shall be fabricated from cold drawn, corrosion-resistant steel, conforming to the composition specified in ASTM F 55 concerning stainless steel material used for surgical implants, except that the carbon content shall be 0.03 percent maximum.

3.1.1 Grain structure. Grain size of the bone plate shall be determined and shall be size 5 or finer in accordance with ASTM F 55.

3.1.2 Inclusion content of material. The nonmetallic inclusion content of the bone plates shall be determined by the JERNKONTORET (JK) chart method in accordance with ASTM E 45, using method "A" or "D". The numbers listed in Table I represent the maximum allowable inclusion ratings.

Table I. Maximum Allowable Inclusion Ratings in Accordance with ASTM E45

	Type A	Type B	Type C	Type D
Thin Series	2	1-1/2	1-1/2	2
Thick Series	1-1/2	1	1	1

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* 3.2 Hardness. Rockwell hardness shall be C30 to C39 when tested as specified in 4.4.2. Rockwell hardness, of any one bone plate, shall not vary by more than 5 points on the Rockwell "C" scale.

3.3 Finish. The completed bone plates shall have a bright (highly reflective) finish, properly electro-polished. Definitions of acceptable surface characteristics shall be as described in ASTM F86.

3.3.1 Passivation. The bone plates shall be properly passivated after completion of all final finishing operations. Passivation shall be accomplished by methods described in ASTM F86.

3.4 Style, design and dimensions. Shall be a one piece, curved bone plate, suitable for implanting into the body for orthopedic repair and internal fixation of bone segments. Bone plates shall be cold drawn from material specified in 3.1 and shall not be formed to shape by stamping operations. Bone plates shall be countersunk holes capable of accommodating the following Bone Screws with 82° Countersunk Head and Pilot Points:

Federal Stock Numbers

6515-366-3010	6515-366-3035	6515-366-3060
6515-366-3015	6515-366-3040	6515-366-3065
6515-366-3020	6515-366-3045	6515-366-3070
6515-366-3025	6515-366-3050	6515-366-3075
6515-366-3030	6515-366-3055	6515-366-3080

Bone plates shall be in accordance with DPSC Drawing 1826 for appropriate size, Federal stock number, and item identification.

3.5 Resistance to boil test, copper sulfate test and saline test. Bone plates shall not show evidence of corrosion when subjected to the boil test as specified in 4.4.1.1; the copper sulfate test as specified in 4.4.1.2; and the saline test as specified in 4.4.1.3.

3.6 Identification marking. Bone plates shall be permanently and legibly marked with the contractor's name or registered trademark, and an indication that the material is stainless steel. Bone plates of foreign (other than U.S.) manufacture shall, in addition to the above, indicate the country of origin. All methods of marking shall be in accordance with ASTM F86.

3.7 Workmanship. Workmanship shall be first class throughout. Bone plates shall be free from defects which detract from their appearance or impair their serviceability, proper functioning, and intended use.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the cold drawn corrosion resistant steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

* 4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table II.

Table II. Sampling Criteria for Examination

Type Defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

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4.2.2 Tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table III.

Table III. Sampling Criteria for Tests

Component	Characteristic	Req't	Test Procedure	Unit Product	Insp Level	AQL
Bone	Grain structure	3.1.1	ASTM F55	1-plate	---	---
	Inclusion content	3.1.2	ASTM E45	*1-plate	S-1	1.0
	Hardness	3.2	4.4.2	1-plate	S-2	1.5
	Dimensions	3.4	----	1-plate	S-2	1.5
	Boil test	3.5	4.4.1.1	1-plate	S-4	0.65
	Copper sulfate test	3.5	4.4.1.2	1-plate	S-4	0.65
	Saline test	3.5	4.4.1.3	1-plate	S-4	0.65

* The unit lot of steel, for which the inclusion content is to be determined, shall not be larger than one heat.

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in construction of the bone plates shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The bone plates shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables II and IV. Examination, however, shall not be restricted to defects listed in Table IV. Any bone plates in the sample containing one or more defects shall be rejected and if the number of defective items in any sample exceeds the acceptance nunber for that sample, the lot represented by the sample shall be rejected.

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Table IV. Classification of Defects

Categories	Defects
Major	
101	Bone plates not of finish specfied.
102	Holes in bone plate not countersunk.
103	Holes in bone plate and outside contours of plates not free from cracks, crevices, burrs, scale, slivers, sharp edges and tool marks.
104	Bone plates not of one-piece construction.
Minor	
201	Bones not permanently and legibly identified as as specified in 3.6.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Corrosion tests.

4.4.1.1 Boil test. Bone plates shall be subjected to the copper sulfate test as described and specified in GG-I-526.

4.4.1.2 Copper sulfate Test. Bone plates shall be subjected to the copper sulfate test as described and specified in GG-I-526.

4.4.1.3 Saline Test. Bone plates shall be scrubbed with soap and warm water, rinsed in hot water, followed by immersion into ethyl alcohol (95 percent) and dried. Bone plates shall then be completely immersed into Normal Saline Solution (U.S.P.) and while immersed, placed into an oven at a temperature of 37 ± 1 degrees Centrigrade for a period of 7 days, removed from the oven and examined for corrosion while still immersed.

4.4.2 Hardness test. Rockwell hardness of the bone plates shall be determined in accordance with method 243 of Federal Method Standard No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One bone plate of the size and style, as specified, constitutes one unit of issue.

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5.1.2 Packaging quantities. The number of units of issue indicated in Table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V. Packaging Quantities

Size	Style	Unit package (units)	Intermediate package (units)	Exterior container (units)
1	1	1	12	600
2	1	1	12	600
3	2	1	12	600
4	2	1	12	600
4	2	1	12	576
5	2	1	12	576
6	2	1	12	432
7	2	1	12	432
8	3	1	12	360
9	3	1	12	288

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

* 5.1.3.2 Preservative. Preservative shall not be applied.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive material.

* 5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method IC-1.

5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, class domestic. Commercial colors are acceptable on intermediate boxes. Closure shall be adequate to prevent spilling of contents under normal handling.

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5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

* 5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box. Fiberboard box shall be waterproofed in accordance with paragraph 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The bone plates are intended for implanting into the body for orthopedic repair and internal fixation of bone segments.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Style and size required (see 1.2).
(c) Selections of applicable level of packaging and packing required (see 5.1 and 5.2).

6.3 This specification covers the items below as they appear in the Federal Supply Catalog. It does not include all sizes or types as indicated by the title nor all types commercially available.

Federal Stock No.	Item Identification
6515-355-9005	BONE, PLATE, 2.25 inch
6515-355-9010	BONE, PLATE, 2.8 inch
6515-355-9015	BONE, PLATE, 3.35 inch
6515-355-9020	BONE, PLATE, 3.9 inch
6515-355-9025	BONE, PLATE, 3.9 inch
6515-355-9030	BONE, PLATE, 4.7 inch
6515-355-9035	BONE PLATE, 5.5 inch
6515-355-9040	BONE, PLATE, 6.3 inch
6515-355-9045	BONE, PLATE, 7.1 inch
6515-355-9050	BONE, PLATE, 7.9 inch

6.4 Supersession data. This specification includes the requirements of Purchase Description No. 6, dated 12 January 1966.

* 6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review activities:
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Project No. 6515-1719

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